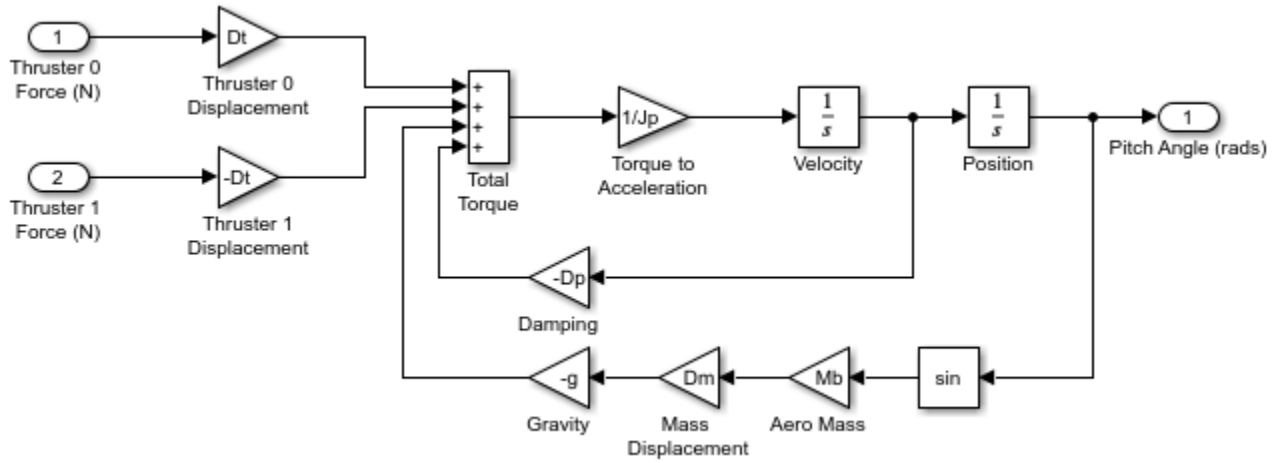
Quanser Aero 2 Recommended Assessment

System Block Diagram Modeling

Note: These results are based on the Aero 2 hardware. The response between different Aero 2 units and the Virtual Aero 2 will vary.

1. Include a screen capture of the subsystem that you created in Step 9 of the System Block Diagram Modeling - Lab Procedure.

   If they followed the lab instructions properly, students should construct a block diagram model similar to that shown below.



2. Attach the scope output of your block diagram model using the default parameters. How well does the model match the hardware response?

   Using the default parameters results in a response similar to that shown here.

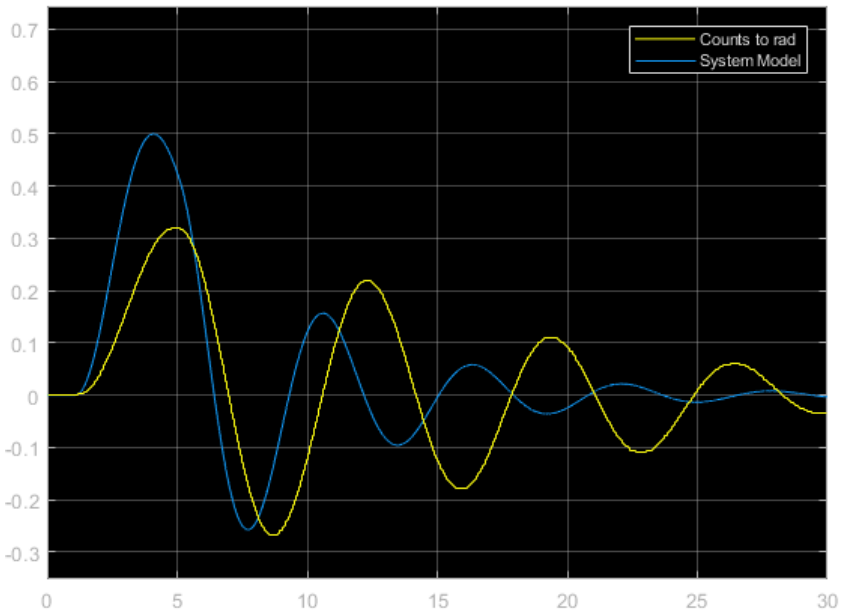


Figure 1 - Model validation response with default model parameters.

While the response is similar in form to the hardware, being a damped oscillation, there remains significant differences in the overshoot, damping and natural frequency of the model.

3. What variables would cause the divergence of the model from the hardware? What adjustments were necessary to better match the hardware behavior?

   Students should observe that the natural frequency of the model is primarily driven by the center of mass displacement, D_m, and that this may vary significantly, being a very small value. The center of mass location changes the stiffness of the system. Thrust and damping can be adjusted to better match the amplitude and decay of the oscillation.

4. Attach the final model result and comment on how well you were able to match the hardware performance. Record the updated values of any parameters you adjusted.

   With some minor adjustment to the parameters, a good approximation of the system response can be achieved using this model. The example shown below used the adjusted values **Dm = 2.6 × 10^{-3} m, Dp = 3.5×10^{-3} N-m/(rad/s)** and **K_{pp} = 2.0×10^{-4} N/(rad/s).** For the Virtual Aero 2, the values would need to be adjusted to **D_m = 1.2×10^{-3} m**, **D_p = 2.5×10^{-3} N-m/(rad/s),** and **K_{pp} = 2.5×10^{-4} N/(rad/s).**

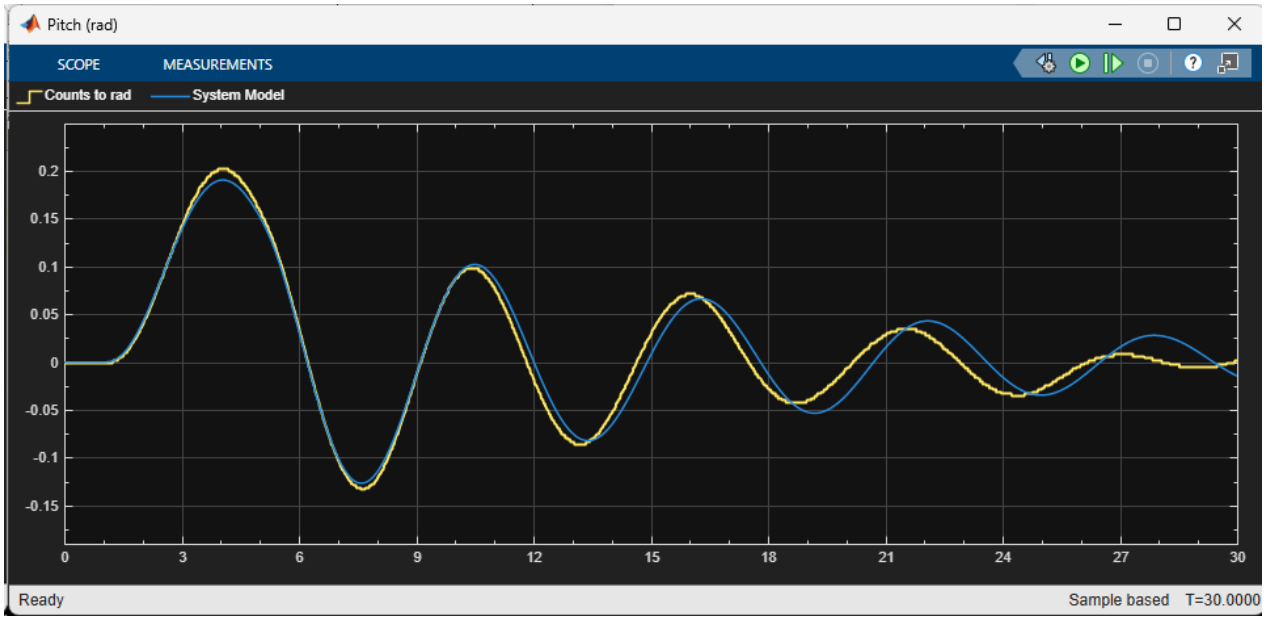


Figure 2 - Sample model validation response from hardware.